D.E → Exact Ordinary

What's behind?

$F(x,y)$

$\alpha_{lk}(t) = (x(t), y(t))$

① $F: \mathbb{R}^2 \to \mathbb{R}$ (Level curves)

$(x,y) \to z = F(x,y)$

② $\alpha_{lk}: [a,b] \leq \mathbb{R} \to \mathbb{R}^2$

$t \to \alpha(t) = (x(t), y(t))$

l_k = level curve

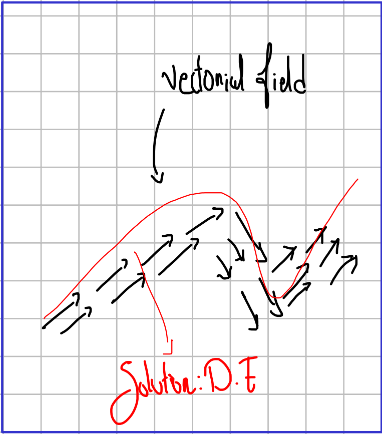
③ $h: [a,b] \leq \mathbb{R} \to \mathbb{R}$

$t \to h(t) = F \circ \alpha_{lk} = k$

compuesto

$h(t) = k$

3 components to study D.E → Exact Ordinary



Behind Camera

$$\frac{dh}{dt}(t) = \nabla F(\alpha(t)) \cdot \frac{d\alpha}{dt}(t)$$

← Product of vectors

vectors

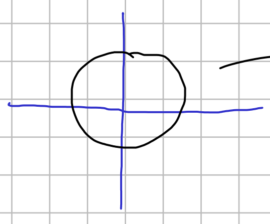
$$\frac{dh}{dt}(t) = \left(\frac{\partial F}{\partial x}(\alpha(t)), \frac{\partial F}{\partial y}(\alpha(t))\right) \cdot \left(\frac{dx}{dt}(t), \frac{dy}{dt}(t)\right)$$

Parametrization

$$\frac{dh}{dt}(t) = \frac{\partial F}{\partial x}(\alpha(x)) \cdot \cancel{\frac{dx}{dt}(t)}^{1} + \frac{\partial F}{\partial y}(\alpha(t)) \cdot \frac{dy}{dx} =$$

$= \emptyset$

Parametrizar



$(\sin(t), \cos(t))$ — $x(t)$, $y(t)$

Parametrizada en x

$(x, \pm\sqrt{1-x^2})$ — $x(t)$, $y(t)$

Exact D.E $= M(x,y) + N(x,y)\frac{dy}{dx} = 0$

- the solution is in the level curve of F which is behind the camera

$F(x,y) = k$

- To find the solution "las derivadas Mixtas" Must be the same

$$\frac{\partial^2 F}{\partial x \partial y} = \frac{\partial^2 F}{\partial y \partial x}$$

(10/April/2026) Class..

Exact D.E: $M(x,y) + N(x,y)\frac{dy}{dx} = 0$

Only if: $\boxed{\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}}$

If F doesn't meet this you can't find F

If this happend $\boxed{\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}}$

Then whe clonclude this

$\boxed{M = \frac{\partial F}{\partial x} \quad y \quad N = \frac{\partial F}{\partial y}}$

you can conclude this

$$F(x,y) = \int M(x,y)\,dx + h(y)$$

constant that depends on 'y'

Problem: $y\cos(x) + 2xe^y + (\sin(x) + x^2e^y - 1)\frac{dy}{dx} = 0$

1) find M and N

$M(x,y) = y\cos(x) + 2xe^y$

$N(x,y) = \sin(x) + x^2e^y - 1$

2) Corroborate $M = \frac{\partial F}{\partial x}$ y $N = \frac{\partial F}{\partial y}$

$\frac{\partial M}{\partial y} = \cos(x) + 2xe^y$

$\frac{\partial N}{\partial y} = \cos(x) + 2xe^y$

3) Exist an F; $\int \frac{\partial F}{\partial x}dx = \int M\,dx$ and $\boxed{\frac{\partial F}{\partial y} = N}$

$$F(x,y) = \int (y\cos(x) + 2xe^y)\,dx + h(y)$$

$$\frac{F(x,y)}{\partial y} = \frac{y\sin(x)}{\partial y} + \frac{x^2e^y}{\partial y} + \frac{h(y)}{\partial y} \qquad \frac{1}{\partial y}$$

$$N = \sin(x) + x^2e^y + \frac{\partial h}{\partial y}$$

$$F(x,y) = y\cdot\sin(x) + x^2e^y - y$$

$$y\sin(x) + x^2e^y - y = K$$